

E-Carry: An Intermittent Pneumatic System for Enhanced Backpack Carrying Comfort

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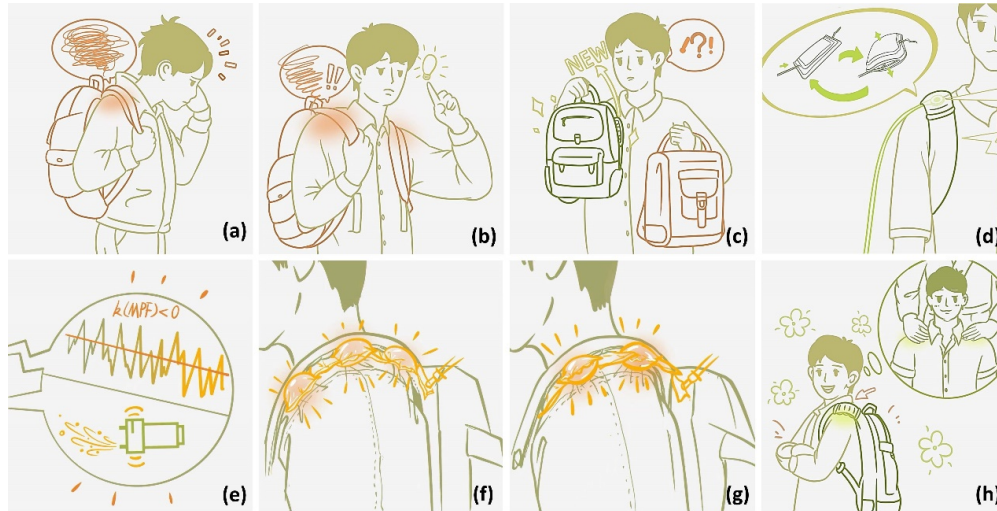


Figure 1: The usage storyboard of E-Carry. (a) Shoulder fatigue from the constant static pressure of a backpack. (b) Proposing a design concept to alleviate the pressure. (c) Change the backpack to E-Carry. (d) Automatic detection of Myoelectric Sensor on E-Carry. (e) The air pump is triggered when the EMG signal detects muscle fatigue (MPF declines). (f-g) Airbag group intermittent inflation. (h) Transform weight-bearing into a positive massage experience, boosting blood flow to relieve fatigue.

Abstract

The backpack that people carry to school or work may lead to shoulder fatigue and pain, due to the constant static pressure it exerts. To address this issue, we propose E-Carry, an intelligent backpack shoulder harness based on the principle of Intermittent Pneumatic Compression (IPC) and background interaction principles (BIP). The E-Carry is designed to utilize intermittently inflating airbags as a dynamic alternating support structure to intervene in shoulder blood flow without distraction and to enhance comfort. In order to explore the effectiveness, in this paper, we conducted a within-subject experiment with twenty-four participants, in which we investigated the blood flow in the shoulder and user experience under three conditions: no weight, traditional backpack, and E-Carry. Results showed that E-Carry significantly improved blood flow, and transformed weight bearing into a positive massage experience by guiding the backpack design through contextual interactions.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices; Haptic devices; Interaction design; Interaction design process and methods; User centered design; Interaction design; Interaction design process and methods; User interface design; • **Applied computing** → Life and medical sciences; Consumer health; • **Hardware** → Emerging technologies; Biology-related information processing; Bio-embedded electronics.

Keywords

Intermittent Pneumatic Compression (IPC), Background Interaction Principles (BIP), Backpack, Shoulder, Blood Flow, Fatigue

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1 Introduction

The backpacks can liberate the wearer's hands [62]. Backpacks worn for extended periods (commuting, school, etc.) [30, 34], require attention to long-term comfort and user experience [30]. Shoulder pressure from backpacks is the main cause of wear-induced fatigue [26, 36, 64]. As the primary load-bearing area [36, 58], the shoulder endures continuous pressure, which can impede blood circulation and reduce metabolic efficiency [10, 41, 43,

49]. Therefore, ensuring ordinary blood flow to muscles is essential for delaying and preventing muscle fatigue [26].

These studies focus on the macrostructure of the backpack [31, 40, 47, 68], attempting to optimize the shoulder pressure distribution through the static structure. For example, studies on localized compression, such as segmented [28, 48] or padded straps [12, 37], aimed to optimize pressure distribution on localized stress surfaces. While Hadid et al. [20] designed ergonomically shaped shoulder straps to promote subclavian artery circulation from the perspective of local blood flow. Still, they fail to address the core issue: Static straps do not accommodate the diverse physiological contours of users [25, 27], potentially resulting in suboptimal localized pressure distribution [20, 38], with the body constantly under constant static pressure [26, 35]. Some studies e.g. latest pneumatic boots for dynamic pressure transfer and feedback to increase blood flow, laminating pneumatic wearable device for managing SCT in cardiovascular disease [53]. In this regard, we aim to propose a dynamic pressure adjustment shoulder strap with intermittently inflated airbags [42, 63] to address the localized discomfort caused by prolonged pressure [2, 3, 26, 63]. The fluid properties of the air in the airbags also help optimize pressure distribution to suit the user's physiological structure.

IPC [42, 63] is a medical technique that provides pressure to the body by inflating and deflating the body for promoting blood circulation [2, 3, 63]. And ensuring normal blood flow to muscles is essential for fatigue [26]. Inspired by this theory, we introduce E-Carry (see in Figure 1), a prototype that utilizes alternating airbags to convert static pressure into dynamic pressure, thereby transforming static load into a dynamic pressure distribution. Enhancing blood flow of the shoulder to relieve fatigue.

Besides, a backpack should not distract the user's attention from act and environment [7]. Therefore, airbags in backpacks should follow the principle of background interaction Principle (BIP) [22]. These predictive interactions, based on sensor data, act as automatic triggers for the device. The primary function of the shoulder strap in a backpack is to distribute weight and pressure, while pneumatic medical products focus on intermittent pressurization, to enhancing blood circulation [2, 3, 63]. Different from other pneumatic information interactive products which requires during usage [60], we activated and deactivated E-Carry by automatically monitoring the fatigue of the shoulder muscles, and alleviate the sense of interaction through the gentle inflation and deflation of airbags, thereby minimizing the need for attention from the user.

Through a within-subjects experiment with 24 participants, we compared shoulder fatigue and user experience under three conditions: no weight, traditional backpack and E-Carry. We addressed three key research questions through controlled trials and user interviews:

- RQ1: Can E-Carry promote shoulder blood circulation during load-bearing conditions?
- RQ2: Whether and how can E-Carry contribute to enhancing the carrying comfort?

The contributions include: i) introducing IPC principles into wearable technology, offering a new approach for relieving fatigue under load-bearing conditions; ii) integrating the BIP to transform the original negative stress of the backpack into a positive one; iii)

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Table 1: Airbag Scores (Focus Group).

Mean	Vertical Bar Airbag	Horizontal Bar Airbag	Lattice Airbag
i	2.33	4.78	4.22
ii	1.33	4.56	4.67
iii	3.22	1.67	3.89
Average	2.29	3.67	4.26

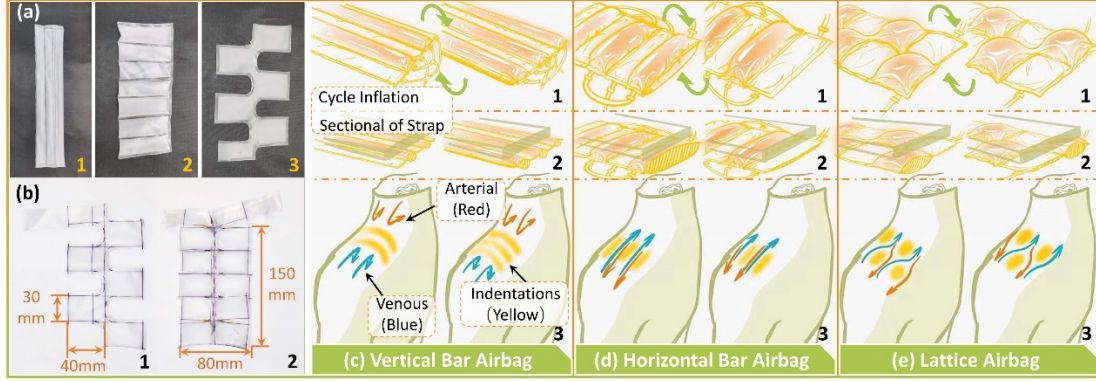


Figure 2: Airbag development process. (a) 1. Vertical bar airbag is distributed from top to bottom of the shoulder straps. 2. Horizontal bar airbag structure follows the course of the shoulder blood vessels and muscles. 3. Lattice airbag is distributed like a checkerboard alternating into two parts. (b) Components and dimensions of airbags. (c) Vertical bar airbag. 1. Airbag inflation status; 2. sectional of the shoulder straps; 3. the direction of blood vessels and muscles, pressure traces and blood flow. (d) Horizontal bar airbag. (e) Lattice airbag.

validating the effectiveness of E-Carry in alleviating shoulder fatigue, enhancing blood flow, and improving overall comfort through user experiments.

2 DESIGN OF E-Carry

2.1 Production of intermittent airbag

To ensure the intermittent airbags meet ergonomic requirements [14, 33, 44], we considered shoulder blood vessels distribution [5, 17, 55]. We developed three types of unit-shaped airbags (Figure 2(a)) and explored longitudinal, transverse and staggered arrangements. The dimensions of airbags are shown in Figure 2(b). All three designs employ a dynamic airbag system with alternating inflation and deflation, named Vertical bar airbag (Figure 2(c)), Horizontal bar airbag (Figure 2(d)), and Lattice airbag (Figure 2(e)). A focus group consisting of seven researchers and two experts evaluated them based on three principles: i) basic function: localizing stability of the shoulder harness; ii) innovative features: reducing obstruction of blood flow, optimizing pressure distribution; iii) comfortable experience: streamlining of structure. Nine participants rated three aspects on a Likert scale of 1 to 5. The average score for each question was calculated and is presented in Table 1. Thus, subsequent development was carried out using the Lattice airbag.

The airbag is made of 0.2mm thick TPU. We tested its airtightness by immersing it in water. During inflation, the pressure felt similar to a single finger's pressure. As shown in Figure 2(b), the shoulder strap width is most appropriate at 80 mm [19], with length was

based on the shoulder trapezius muscle covered by the compression zone (from the apex of the shoulder strap to the clavicle). Air holes are mounted in the upper left and upper right extensions of each set of airbags, avoiding body contact (See in Figure 2(b) and Figure 3(b)). The inflation settings favoring blood flow refer to the relevant literature [42], and the airbag inflation time is less than 0.4 seconds with a pressure of approximately 100 mmHg and a duration of 3 seconds.

2.2 Interaction design and Function Realization

The system is based on BIP, detects and manages shoulder muscle fatigue from heavy loads using the muscle electrical signal sensing module (190Hz includes dominant signal components [32], with muscle fatigue characterized by increased amplitude and a shift from high to low frequency spectra [51]). This module, affixed to the shoulder's load-bearing area and collects real-time electromyographic signals. Individual's shoulder fatigue can be identified by the slope (k) of a linear fit to the Mean Power Frequency (MPF, a key parameter for fatigue monitoring), where a $k_{(MPF)} < 0$ represents muscle fatigue. The formula used is the integral of the product of the frequency (f) and the corresponding power spectral density function $P(f)$, divided by the integral of $P(f)$. The equation used is as function (1):

$$\frac{\int_{f_1}^{f_2} f \cdot P(f) df}{\int_{f_1}^{f_2} P(f) df} \quad (1)$$

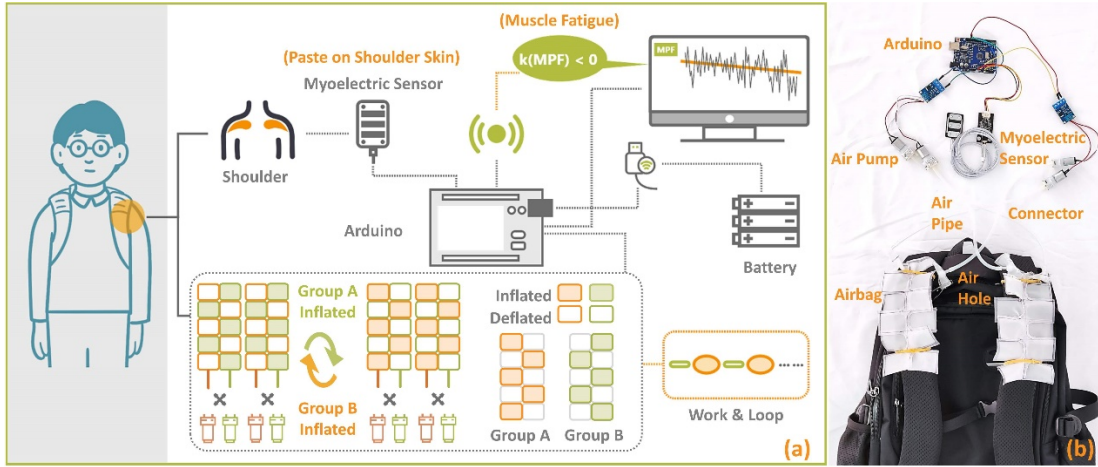


Figure 3: E-Carry system. (a) The interaction principle of E-Carry. (b) E-Carry's hardware.

According to BIP, the myoelectric sensor is attached near the shoulder strap. If the MPF linear regression slope is negative, it indicates muscle fatigue, activating the E-Carry system (See Figure 3(a)). The airbags then inflate for 3 seconds, then deflate for 3 seconds in a continuous cycle, with alternating sets to ensure one set is always inflating. This supports pressure redistribution, while the deflated airbags enhance blood flow in the shoulder area (See Figure 3(b)). Myoelectric sensors leave the skin (take off the backpack), E-Carry stops.

3 Experiment

3.1 Participants

24 participants (12 males, 12 females), aged 20 to 26 ($M = 24.17$, $SD = 1.37$), with a mean height of 171.85 cm ($SD = 7.68$) and a mean weight of 67.17 kg ($SD = 12.50$), were recruited via online announcements and offline mobilization among alumni. There were no restrictions on age, gender, or major, but participants were required to regularly use backpacks. Prior to testing, participants were informed of the study protocol and potential risks, and signed a consent form. Individuals with a history of upper limb or back musculoskeletal disorders were excluded. To minimize the impact of clothing on weight bearing, participants were asked to wear loose fitting t-shirts.

3.2 Experimental Setup

To ensure scientific validity, the experiment was conducted in a controlled laboratory environment at 25°C. The following experimental equipment was used: an Edison®backpack (0.7 kg, China's largest online shopping platform with 50,000 sales.), Laser Doppler flowmetry (Biopac, 2000 Hz) [4, 23, 41], and an LDF 100C Probe (TSD140 series) were used to measure the blood flow velocity at the participant's shoulder. The study has been reviewed and approved by the Ethics Committee of the Beijing Institute of Technology (Reference number: BIT-EC-H-2024002). All experimental data and written materials are stored on a password-protected server accessible only to the study team.

3.3 Procedure

The experiment includes three conditions with weight and backpack type as independent variables and skin blood flow as the dependent variable: No weight (baseline), 5 kg with a traditional backpack, and 5 kg with E-Carry. The 5kg with a traditional backpack condition refers to a backpack without any shoulder modifications, where weight is added to reach a total of 5kg to simulate carrying a load in daily situations. In contrast, 5kg with E-Carry refers to the same backpack condition, but with the addition of the E-Carry system at the shoulder straps. Additionally, the experimental environment as a controlled variable, remains consistent across all conditions. Participants were tested under all three conditions, each lasting 15 minutes, with a 15-minute break between tests during which a questionnaire was administered. To minimize order effects, the no weight condition was always tested first, followed by counterbalanced testing of the traditional backpack and E-Carry. The experimental equipment is shown in Figure 4(a-d).

The experimental procedure was divided into 4 steps. **Step 1:** Before the experiment began, participants first took a 15-minute break (Figure 4(e)) to initialize their individual shoulder pressure. **Step 2:** The sensors were fixed to the participants. We cleaned their skin with alcohol (Figure 4(f)), and the LDF probe (Figure 4(g)) was attached below the shoulder girdle, and between the first thoracic spine and the second thoracic spine [41]. Taping the probe (Figure 4(h)). During each experimental condition, blood flow data collection was initiated throughout the process. **Step 3:** Conducting experiments under three conditions. Figure 4(j) shows the no weight condition. Figure 4(k) represents the traditional backpack condition, and Figure 4(l-m) is the E-Carry condition, where airbags are inflated alternately. Each of the condition took 15 minutes. **Step 4:** Scales and interview. RPE scale [59] (Figure 4(i)(n)) was collected with backpack from participants before and after each experimental condition. After the traditional backpack and E-Carry experimental conditions, participants were asked to complete the IMI [15, 50], and UEQ [52, 54] questionnaires (Figure 4(o)) and take an interview.

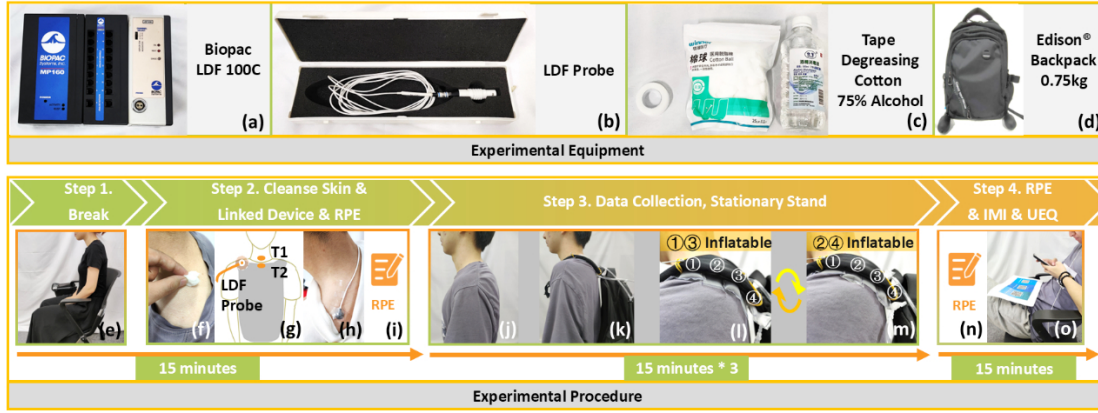


Figure 4: Experimental equipment and processes. (a-d) Experimental equipment. (e-o) The experimental procedure was divided into 4 steps. (e) Rest to restore normal blood flow. (f) Cleaning the skin. (g) LDF probe is taped below the shoulder strap, between the first and second thoracic vertebrae. (h) Pasting probe. (i) Filling out RPE. (j) No weight. (k) Traditional backpack. (l-m) E-Carry. (n-o) Filling out RPE, IMI and UEQ scales.

3.4 Data collection and analysis

We measure shoulder blood flow using Clifford P. Mao's method [41]. Subjective fatigue is assessed using the Borg RPE scale [59] (Likert scale 6 to 20), subjective initiative is assessed using the IMI [15, 50] (Likert scale 1 to 5), and user experience is assessed using the UEQ [52, 54] (Likert scale 1 to 5).

Quantitative data were analyzed using SPSS software, starting with the Shapiro-Wilk test for normality, and non-parametric tests were to data that did not follow a normal distribution. Qualitative data were collected through semi-structured interviews, on participants' perceptions. Observational data were collected throughout the experiment by researchers. Qualitative data was conducted, with six researchers independently coding the data and discussing to identify three main themes in section 4.2.1, 4.2.2 and 4.2.3.

4 RESULT

4.1 Quantitative results

4.1.1 E-Carry significantly enhanced shoulder blood circulation under load conditions (RQ1). As shown in Figure 5(a), the results showed that compared to the no weight ($M=26.23$, 95% Confidence Interval (CI), 19.65-32.81, $SD=15.58$), traditional backpack ($M=12.66$, 95% CI, 10.26-15.05, $SD=5.68$) was 51.73% decreased in average blood flow velocity, while E-Carry ($M=29.59$, 95% CI, 22.83- 36.34, $SD=15.99$) resulted in a 12.81% increase. Both E-Carry ($P<0.001$) and no weight ($P<0.001$) were significantly higher than the traditional backpack condition. E-Carry had a higher shoulder blood flow rate than no weight, but not significantly different, which suggests that the shoulder blood flow with the E-Carry is consistent with that of the no weight.

4.1.2 RPE Results (RQ2). The results (see Table 2) of the Friedman Test showed that there was no significant difference in shoulder fatigue levels between no weight and E-Carry conditions. However, shoulder fatigue levels in traditional backpack condition were significantly higher than those in no weight and E-Carry conditions. This indicated that using E-Carry helps to reduce the shoulder fatigue

associated with carrying 5kg load, making it comparable to the shoulder fatigue level experienced during 15 minutes of standing without any load.

4.1.3 IMI Results (RQ2). The IMI scale statistics (shown in Figure 5(b)) indicated that E-Carry yields significantly higher levels of Interest/Enjoyment ($M = 5.66$, $SD = 1.48$), Perceived Choice ($M = 6.15$, $SD = 0.85$), and Value/Usefulness ($M = 4.73$, $SD = 1.69$) compared to using traditional backpack, with $p<0.001$. The scores for traditional backpack were: Interest/Enjoyment ($M = 2.97$, $SD = 1.58$), Perceived Choice ($M = 3.81$, $SD = 1.95$), and Value/Usefulness ($M = 3.56$, $SD = 1.72$) were around the median of the scale.

4.1.4 UEQ Results (RQ2). As shown in Figure 5(c), the UEQ results indicated that E-Carry significantly outperforms the traditional backpack in the dimensions of Pragmatic Quality ($M = 0.99$, $SD = 0.81$), Hedonic Quality ($M = 1.92$, $SD = 0.52$), and Overall Experience ($M = 1.46$, $SD = 0.57$), with $p < 0.001$. The traditional backpack received lower scores in Pragmatic Quality ($M = 0.02$, $SD = 0.82$), Hedonic Quality ($M = -1.25$, $SD = 1.20$), and Overall Experience ($M = -0.62$, $SD = 0.85$). This suggests that E-Carry significantly enhances the user experience compared to the traditional backpack.

4.2 Qualitative Result

In this section, we explain how E-Carry enhances usage motivation and user experience based on participants feedback, addressing RQ2.

4.2.1 The IPC-based intermittent pneumatic shoulder strap effectively promotes dynamic pressure distribution. The alternating distribution of airbags and the dynamic inflation process prevent sustained pressure. The E-Carry's airbag inflation and deflation mechanism adjusts support based on weight distribution, reducing shoulder discomfort. P19 added, "The inflating straps make my shoulders feel more comfortable, without constant squeezing." P13 emphasized, "Inflation distributes weight effectively, reducing back pressure and enhancing comfort," and P8 stated, "It reduces fatigue

Table 2: Mean (SD) and Friedman Test Results.

	Pre	Post	P
No Weight	8.52 (1.58)	11.80 (2.20)	.001***
Traditional Backpack	8.64 (2.04)	15.00 (2.10)	.001***
E-Carry	8.68 (1.44)	12.60 (1.78)	.001***
P (No Weight vs. Traditional Backpack)	.815		.001***
P (No Weight vs. E-Carry)	.374		.07
P (Traditional Backpack vs. E-Carry)	.936		.001***

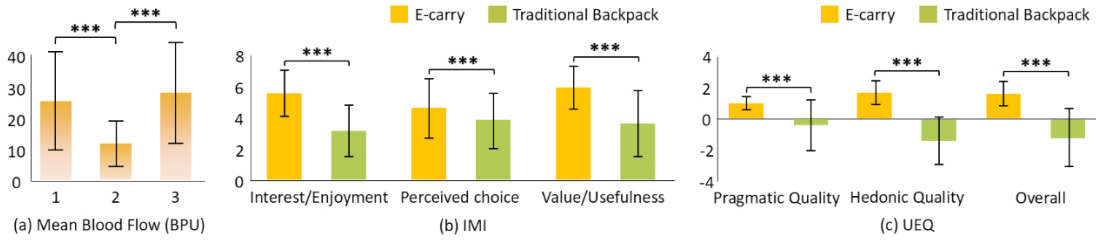


Figure 5: Data statistics. (a) Mean blood flow (see Table 3 for metadata in appendice), Biopac LDF100C unit: BPU. 1. No weight (baseline), 2. traditional backpack 3. E-Carry. (b) IMI results. (c) UEQ results.

and avoids constant pressure on my shoulders.” The airbag shoulder straps also distribute pressure more effectively. P6 noted, “The E-Carry’s airbag feels like an active point moving across different areas.” P24 observed, “The backpack feels lighter with inflated airbags, and my shoulders bear less weight when they deflate.”

Modular intermittent inflatable airbags automatically fit different people’s shoulder structures. Participants noted that there was no significant concentration of pressure on the shoulder with the E-Carry. This is because the distribution of airbags adapts more effectively to the curvature and area of the shoulder. P6 observed, “The traditional backpack force area was concentrated only on the top of the shoulder, the E-Carry force extends towards the back of the shoulder.” P10 added, “The skin on my shoulder appeared a regular pink color after the test, and a traditional backpack would leave me with lines of strangulation on my shoulders.” P18 noted, “E-Carry provides a more resilient muscle compression experience.”

4.2.2 E-Carry with BIP facilitated both physical and mental experiences. The unconscious experience of massage promotes mental relaxation. E-Carry’s airbags anticipate fatigue and provide a massage-like experience to eliminate mental fatigue. P24 added, “The pressure points keep moving, making the shoulders feel lighter.” P23 likened it to, “Sitting in an automatic massage chair that massages your shoulders.” Additionally, P1 added, “When using the E-Carry, I don’t worry at all about it interfering with my commute.” P4 added, “Automatic implementation of new features without interfering with the use of existing features for better satisfaction.” Air pump noise in the use of the backpack environment, the impact on the use of the experience is small; backpack use scenarios focused on the road, the environmental noise of these public places can mask most of the air pump noise, such as buses, subways, roads and shopping malls (P6, P10, P11, P18 and P22).

Converting negative carrying pressure into positive massage pressure. The E-Carry system enhances physical comfort

by converting load weight into a massage-like pressure through intermittent air bladders. P17 described it as, “I had chronic shoulder fatigue and it massaged me automatically very quickly after use.” P20 described, “The weight of the backpack is evenly distributed across the air bladders, almost like a massage.” P11 added, “Rhythmic inflation and deflation creates soothing pressure and a slight warmth in the shoulders after use.” P8 emphasized, “The massaging effect of the air bladder helps reduce the discomfort of prolonged compression. Traditional shoulder strap padding tends to lose its cushioning effect over time.”

4.2.3 Design Opportunities. It was recommended that the frequency of intermittent inflation and the amount of inflation should be adjusted according to personal preference (P1, P3 and P15). Adjust the size of the airbag to the width of the backpack straps to accommodate more users (P4, P8). Extend and widen the airbag and apply it to other backpack areas (P12). Would like the Myoelectric Sensor to be a little smaller and the lines to be fused into the fabric in the product (P7, P16 and P22). Existing Myoelectric Sensor are too hard and cold, would like to have Myoelectric Sensor that do not touch the skin (P4, P10 and P20).

5 DISCUSSION

This paper introduces E-Carry, an interactive shoulder strap system using IPC and BIP to reduce shoulder discomfort from heavy loads. E-Carry improves blood circulation during load-bearing (RQ1) and offers better user experience and greater willingness to use compared to traditional backpacks (RQ2). Participants highlighted its ability to transform negative pressure into a comfortable, massage-like effect without deliberate operation.

5.1 Utilizing Distributed Airbags to Enhance the Uniformity of Pressure Distribution

The pneumatic structure of E-Carry improves comfort by alternating airbag distribution. This design reduces pulling and sagging, ensuring even pressure and enhanced comfort during use. The reason for this is that air has fluid characteristics that automatically adapt to the different physiological curves of the shoulder. Consistent with prior research [1, 29, 57, 60, 66], redistributing shoulder pressure alleviates fatigue. Future studies could explore modular and distributed shoulder straps for adaptive fit and better user experience [8, 60, 66].

5.2 Utilizing Medical Principles to Promote Blood Circulation

Compared to previous pneumatic products, the E-Carry not only cushions the pressure on the shoulders, but also converts the original negative pressure of the backpack into positive pressure. The integration of medical rehabilitation principles into wearable technology reflects a trend toward combining health functionality with daily usability [21, 24, 46]. By bridging medical rehabilitation and daily use, it enhances health outcomes and user comfort. However, consumer products must address user compliance [9, 39, 67], device accessibility [56], and over-reliance on technology must be addressed in consumer products [11]. Traditional backpack restrict blood flow, causing fatigue and pain [6, 13, 20, 41]. Our experiments showed a significant increase in blood flow rate in the shoulders with E-Carry, and not significantly different from no weight condition. E-Carry maintains ordinary blood flow, facilitates the elimination of acid [26], and reduces tissue hypoxia risks. Future studies should expand to diverse populations and real-world scenarios to explore its broader application [16, 18, 34, 45].

5.3 Utilizing BIP to Optimize the Allocation of Attention

On the wearable side, BIP can help the brain process non-urgent information, keep in-the-moment activities safe, prevent hidden dangers and enhance the user experience. According to Maslow's Hierarchy of Needs [65], during a backpacking activity, the user needs to pay more attention to the activity and its surroundings [7] as it is likely to involve safety. The second is the experience, but this does not mean that the experience is not important, a bad experience is likely to be a health alarm, just like the muscle soreness brought by backpacking has the potential to evolve into a disease [48, 61, 64, 69]. Therefore, experiences that are safety hazards need to be addressed through BIP based on ensuring safety of movement in the moment.

6 LIMITATIONS AND FUTURE WORK

The relatively small sample size and time period of the experiment may limit its generalizability. Future studies will expand the sample size and population diversity. In the selection of airbags, we have currently employed a focus group approach. Future research will optimize the design and usage strategy of the airbags through the incorporation of quantitative methods such as blood flow detection. Additionally, the current E-Carry is in the prototype stage, requiring

sensors to make direct contact with the skin to assess fatigue levels. Future research will focus on exploring better ways to integrate medical technologies into wearable products.

7 CONCLUSION

This paper applied IPC and BIP to a dynamic shoulder strap design, using distributed airbags to enable dynamic pressure distribution. A controlled experiment with 24 participants showed that E-Carry is effective in improving shoulder blood circulation and reducing fatigue physiologically and psychologically. The results indicated that E-Carry effectively mitigated the negative impact of prolonged backpack carrying on shoulder blood flow, and significantly enhanced participants' usage motivation and user experience compared to traditional backpack. The E-Carry's automated, mindless interaction design also reduced the user's operational burden, allowing the pack to provide comfort and health benefits without interrupting the user's attention. This paper provides a dynamic approach to alleviating shoulder fatigue in backpack shoulder straps, and offers insights into the integration of medical principles with wearable products.

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A APPENDICE

Table 3: Metadata for Blood Flow Experiments.

Subject	No Weight (Base Line)			Traditional Backpack			E-Carry		
	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean
P1	419.921	6.713	21.591	304.260	2.441	8.074	1049.804	1.831	21.076
P2	85.144	5.493	17.764	161.132	6.713	15.497	464.782	7.019	23.512
P3	105.895	10.681	46.677	449.218	7.934	20.964	996.398	10.070	31.154
P4	216.979	10.070	21.290	128.784	2.136	18.561	248.107	0.305	31.729
P5	164.794	7.324	20.210	90.332	3.051	12.938	922.851	2.746	25.598
P6	162.353	11.596	24.592	292.358	0.915	8.736	426.94	1.831	14.045
P7	79.040	5.187	12.289	48.217	3.356	8.980	190.429	0.305	14.083
P8	50.354	6.713	12.290	41.198	0.610	4.378	144.958	0.610	10.002
P9	211.486	6.713	25.53	239.868	3.967	10.257	835.266	1.831	47.151
P10	262.451	9.765	20.846	129.699	6.103	20.582	235.290	7.629	24.877
P11	352.783	6.408	12.300	87.890	3.356	8.139	398.254	3.967	27.998
P12	484.008	17.395	38.458	150.451	3.967	22.610	235.290	7.629	24.877
P13	71.716	8.239	19.518	165.405	2.746	12.096	498.352	0.305	17.778
P14	272.521	21.362	85.906	246.887	4.272	15.831	658.569	9.46	77.943
P15	188.903	10.070	21.503	359.000	2.136	10.051	150.146	6.713	28.408
P16	101.013	11.291	22.814	86.669	7.019	20.526	364.990	4.577	14.435
P17	55.541	18.920	31.236	48.828	4.882	15.661	222.473	5.493	26.332
P18	226.745	7.934	30.356	375.360	2.746	7.954	614.624	0.305	57.045
P19	52.185	7.324	19.773	35.705	1.525	5.318	345.458	2.746	31.315
P20	303.955	11.596	42.411	45.471	3.051	7.030	304.87	3.356	24.263
P21	123.596	9.155	16.320	83.923	3.967	17.899	132.141	6.713	19.492
P22	117.187	10.681	25.432	108.642	4.882	18.015	370.788	2.746	56.424
P23	88.195	7.629	13.545	81.176	1.525	5.623	448.303	2.136	40.372
P24	82.720	9.765	26.891	72.242	3.356	8.009	222.778	6.408	20.232